# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I* L.Charan Saireddy *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

| Criteria | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | 2 | 2 | 2 | 2 |
| Explanation | 25% | 2 | 2 | 2 | 1 |
| Application | 20% | 1 | 1 | 1 | 1 |
| Organization | 15% | 1 | 1 | 1 | 1 |
| Accuracy | 10% | 1 | 1 | 1 | 1 |
| Clarity | 5% | 1 | 1 | 1 | 1 |

**Total Marks** 20 /40

***Signature of the Faculty***

## Short Answer

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

**9.**A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

**10.**A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

**RUBRICS FOR CALCULATION**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |

| | | | | | |
|---|---|---|---|---|---|
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organization | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |

No plication rrect

CO1 (Short)

ASSESMENT TOOL - 1

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40/40

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1) Total Number of Segments and frames

Total file Size in byte: 150 MB = $150 \times 10^6$ byte

= 150,000,00 byte

Segment Size : 1500 bytes

Number of Segments:

$$\frac{150,000,000}{1500} = 100,000 \text{ segments}$$

- Number of frames: Since each Segment maps to exactly one frame, there are 100,000 frames

2. Total transmission Overhead.

- overhead per frame : 40 byte
- Total Data link layer overhead : 100,000 x 40 byte = 4,000,000 byte = 40 MB.

3. Total transmission time.

- Total Size of a frame: 1500 bytes (Segment) + 40 bytes (Headers/trailers) = 1540 bytes

- Total bits transmitted.

  100,000 frames × 1540 bytes/frame × 8 bits/byte

  = 1,232,000,000 bits.

- Network bandwidth 50Mbps = 50,000,000 bit/s

- Transmission time:

$$\frac{1,232,000,000}{50,000,000} = 24.64 \text{ second.}$$

LAYER RELIABILITY MECHANISM

1) TRASPORT LAYER: Ensure reliable end to end communication b/w applications. It uses Mechanism like acknowledgments (ACKs) Sequence Number to resemble out of order data segments, and retransmission protocols (eg. TCP) to recover lost segments.

- Data link Layer: Ensure reliable Communicati across a single physical link between adjacen nodes. It handles error detection and correctio by computing a checksum or cyclic Redundancy check (CRC) placed in the fram trailer to identify and discard corrupted frames.

visualizing either data and trailers data.

Visualizing Data Encapsulation & overhead

When data travels down the OSI stack, header and trailers wrap around the original application data. There is how the 1500-byte Data link frame is composed relative to the 1500 - byte Segment

2. Network efficiency & Goodput Analysis

- Protocol Efficiency: Measure the percentage of the transmitted bits that belong to the actual payload.

$$\text{Efficiency} = \frac{\text{Payload Size}}{\text{Total frame Size}} = \frac{1500 \text{ byte}}{1594 \text{ byte}} \approx 97.4\%$$

- Good put (Effective Throughput).

The actual rate at which useful application data is delivered over the network.

Good put = Bandwidth x Efficiency.

= 50 Mbps x 0.974

≈ 48.7 Mbps.

39 1) Fragmentation & overhead calcula

- Total Packet size: 12,000 bytes (Excl
the original 20 byte IP header + 11
byte of data playload).
- Maximum transmission unit (MTU): 1500 by
- Maximum payload per fragment: 1500
byte (MTU) - 20 byte (IP header) = 1480 bytes
- Note: 1480 is divisible by 8, which satisfie
the required fragment offset alignment.

Fragment Breakdown:

To transmit 11980 byte of data playload.

- Fragments 1 to 8: Each carries 1480 byte of payload (8 x 1480 = 11,840 bytes).
- Fragment 9: carries the remaining payload (11980 - 11840 = 140 bytes).
- Total fragments: 9.

Total Header Overhead Introduced:

Since the original packet already included one 20-byte header, the extra overhead comes from the 8 newly created fragment headers.

Alternative
all fragm
160 bytes
header

Overhead Introduced = 8 x 20 bytes = 160 bytes

Alternatively, the total header weight accross all fragment is 9x20 = 180 bytes, which is 160 bytes more than the original 20 byte header.

2. How the Network Layer Ensures Successful Deliver

The IPV4/IPV6 Network Layer uses Specific field inside the IP header at the destination host to reassemble the fragment correctly:

- Identification (ID) field:

All fragments originating from the same initial packet share the exact same unique ID number this allows the destinati to group them together and separate them from other packets.

- Fragment offset field: Indicates where the specific fragments payload belong relativ to the start of the original data. Payload (Measured in unit of 8 bytes).

This allows the destination to sequence and stitch the pieces together in the correct order, even if they arrive out of sequence.

- More fragment (MF) Flag:
- Set to 1 for fragments 1 through 8, telling the receiver that more pieces are coming
- Set to 0 for the final (9th) fragment, signaling to the receiver that the packet is complete and ready for reassembly.

If any single fragment is lost or dropped in transit, the network layer cannot completely reassembly and the entire packet is discarded, requiring an upper-layer protocol (like TCP) to trigger a retransmission.

# Calculating the total time (End to End delay)

When we talk about end to end delay we are measuring the total amount of time it takes for a message (a packet) to travel all the way from the starting device to the destination device.

To find the total time, we simply add together three main parts.

1) Processing Delay: The time the computer takes to pack and prepare the data at every layer

2) Transmission Delay: The time it takes to push the data out onto the wire or into the air.

3) Propagation Delay: The time it takes for the signal to travel through the physical wire or air over a distance.

Step 1: Adding up Internal work (Processing Delay)

Inside the computer the data travels down through 5 different layers. Each layer takes a tiny amount of time to do its job

- Application layer: 4ms.
- Transport layer : 3ms.
- Network layer : 5ms
- Data link layer : 2ms
- physical layer : 1ms.

Time = 4ms + 3ms + 5ms + 2ms + 1ms = 15ms

Step 2: Adding the travel times.

Now that the data is ready, we add the time it takes to get onto the network and travel across it.

- Transmission Time (getting onto the wire) : 12ms
- Propagation time (Moving through the distance) : 18

Step 3: the final Sum.

Now we add everything together to get the complete Journey time.

Total time = 15ms (processing) + 12 ms (Transmission) + 18ms (Propagation)

= 15 + 18 + 12 =

= 45 :

∴ Total time = 45ms.

Layer Contribution to the Communication process.

Each Mentioned layer plays a specific role in preparing moving and delivering the packet:

- Application layer (7ms): Interacts directly with the Software application. It formats, prepares and initiates the data payload for transmission (eg., HTTP, FTP).
- Transport layer (3ms): Manages end to end communication, segmentation and reliability. It adds source/destination port number and handles flow or error control (eg. TCP, UDP).
- Network layer (5ms): Handles logical addressing and routing. It add IP addresses to the packet and determine the best physical path across networks.
- physical layer (1ms): convert the binary data frames into physical signals. and transmit them onto the network Medium.

9) To transfer 80MB of useful Data. 52,632 frames are required, resulting in 4,210,560 bytes of total overhead and a transmission efficiency of 95%. Here is the step by step breakdown of the calculations.

1) Calculate useful Data per frame
- Total frame size : 1600 bytes.
- protocol overhead per frame : 80 byte.
- useful data (Payload) per frame
  -> 1600 byte - 80 bytes
  = 1520 bytes

2) Total Number of frames required.
- Total useful Data to transfer : 80MB = $80 \times 10^6$ bytes = 80,000,000 byte (using standard Metric decimal unit for network data size.
- Number of frames.

$$\frac{80,000,000}{1520 \text{ byte/frame}} \approx 52631.58 \approx 52632 \text{ frames}$$

Total overhead transmitted

Total overhead: 52,632 frame x 80 byte / frame
= 4,210,560 byte (or ~4.21MB).

4. Transmission Efficiency.

$$\text{Efficiency} = \frac{\text{useful data per frame}}{\text{Total frame Size}} \times 100\%$$

$$\text{efficiency} = \frac{1520 \text{ bytes}}{1600 \text{ bytes}} \times 100\% = 95\%.$$

Impact of protocol overhead on Network Performance.

Protocol overhead (headers / trailers added by OSI layer like TCP, IP, Ethernet) affects network performance in the following ways

- Reduces Available Throughput (Goodput): part of the network's bandwidth is consumed by header rather than user data, lowering the actual useful data transmission rate.
- Higher overhead: Lead to lower efficiency, slower data transfer rate. and increased congestion.

- Lower overhead: Maximizes data throughput but can increase risk if necessary control/error-checking features are reduced